



Spring Testing

Understanding Mockito

Mockito

Mockito is useful for testing components in isolation without relying on their actual implementations or external dependencies.

Mockito can be used for:

- **Mocking**: Creating mock objects to simulate the behavior of real objects.
- **Stubbing**: Defining the behavior of mock objects when specific methods are called.
- **Verification**: Checking if certain methods were called on mock objects



Mockito Methods

1. Creating Mock

with `@Mock` Annotation

with `Mockito.mock(Classname)`

2. Stubbing Mock

`when(T methodCall)`: Specifies a method call to be stubbed.

`thenReturn(T value)`: Sets a return value for a stubbed method call.

`thenThrow(Throwable... throwables)`: Throws an exception when a stubbed method call is made.

Mockito Methods

3. Verifying Methods

`verify(T mock)`: Verifies that a method was called on a mock.

`verify(T mock, VerificationMode mode)`: Verifies that a method was called with a specific verification mode (e.g., times, never).

VerificationMode modes:

- `times(int wantedNumberOfInvocations)`: Verifies the exact number of invocations.
- `never()`: Verifies that a method was never called.
- `atLeastOnce()`: Verifies that a method was called at least once.
- `atLeast(int minNumberOfInvocations)`: at least called x times
- `atMost(int maxNumberOfInvocations)`: at most called x times
- `only()`: Verifies that no other method was called on the mock.

Mockito Methods

4. Argument Captor:

It allows you to capture arguments for further assertions or validation, making it useful when you want to inspect the arguments that were passed to a method call.

```
// Call the method to be tested
employeeService.saveEmployee(employee);

// Create an ArgumentCaptor for Employee
ArgumentCaptor<Employee> employeeCaptor = ArgumentCaptor.forClass(Employee.class);

// Verify the save method was called and capture the argument
verify(mockRepository).save(employeeCaptor.capture());

// Retrieve the captured argument
Employee capturedEmployee = employeeCaptor.getValue();

// Perform assertions on the captured argument
assertThat(capturedEmployee.getName()).isEqualTo("John Doe");
assertThat(capturedEmployee.getAge()).isEqualTo(30);
```

